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**Department of Community Health  
COMH 603 EPIDEMIOLOGY 1  
Final Exam  
Friday 1st December, 2006**

Instructions to Students

1. You have 3 hours in which to complete this exam
2. Fill in your name at the top of each sheet of paper
3. Calculators may be used
4. Show your workings for each calculation
5. Circle the single correct answer for multiple-choice questions, otherwise follow the instructions given
6. Each correct multiple-choice response will be given one mark; marks for other questions are as shown.

Questions 1-3 refer to the table below, which shows data on cause of death by smoking habit among 34,440 participants in a large cohort study following doctors.

**Death rate of men by cause of death and smoking habit**

| Cause of death                | Death rate per 100,000 men per year |                                      |       |     |             |
|-------------------------------|-------------------------------------|--------------------------------------|-------|-----|-------------|
|                               | Non-smokers                         | Current smokers (cigarettes per day) |       |     | All smokers |
|                               |                                     | 1-14                                 | 15-24 | 25+ |             |
| Lung cancer                   | 10                                  | 78                                   | 127   | 251 | 140         |
| Ischaemic heart disease (IHD) | 413                                 | 608                                  | 652   | 792 | 669         |

1. Calculate the rate difference and the rate ratio for each level of smoking and each cause of death, using non-smokers as the comparison group. (Fill in the table below). (8 marks)

| Cause of death | Rate difference/100,000 men/yr |       |     |     | Rate ratio |       |      |     |
|----------------|--------------------------------|-------|-----|-----|------------|-------|------|-----|
|                | 1-14                           | 15-24 | 25+ | All | 1-14       | 15-24 | 25+  | All |
| Lung cancer    | 68                             | 117   | 241 | 130 | 7.8        | 12.7  | 25.1 | 14  |
| IHD            | 195                            | 239   | 379 | 256 | 1.47       | 1.58  | 1.92 | 1.6 |

2. Which disease is **most strongly associated** with smoking? (1 mark)  
Lung cancer

3. Which disease has the largest **number of deaths attributable** to smoking? (1 mark)  
IHD

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Questions 4-6 refer to the table beneath:

Nurses working in radiology and gynecology were followed for 10 years to assess the risk of developing thyroid cancer.

| Department                | Radiology | Gynecology |
|---------------------------|-----------|------------|
| Number of nurses          | 50        | 50         |
| Person-years of follow-up | 500       | 500        |
| Cases of thyroid cancer   | 6         | 2          |

4. The incidence rate of thyroid cancer among radiology nurses is (1 mark):
- A. 167 per 1000 person years
  - B. 94 per 1000 person years
  - C. 13.6 per 100 person years
  - D. 12 per 1000 person years
  - E. 12 per 100 person years
5. The odds of developing thyroid cancer among gynecology nurses is (1 mark):
- A. 0.004
  - B. 0.040
  - C. 0.042
  - D. 48.0
  - E. 498
6. The risk ratio of thyroid cancer among radiology nurses compared with gynecology nurses is (1 mark):
- A. 0.67
  - B. 1.0
  - C. 1.5
  - D. 2
  - E. 3
- e7. A new treatment for thyroid cancer is introduced, so patients live on average twice as long after diagnosis. Which of the following is likely to be true once the treatment is widely available (1 mark)?
- A. The incidence of thyroid cancer will decrease.
  - B. The prevalence of thyroid cancer will decrease.
  - C. Both the incidence and the prevalence of thyroid cancer will decrease.
  - D. The incidence of thyroid cancer will increase.
  - E. The prevalence of thyroid cancer will increase.
8. Which of the following is *not* used to define causality in an exposure-disease association (1 mark)?
- A. Consistency
  - B. Strength of association
  - C. Sensitivity
  - D. Specificity
  - E. Temporality

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Questions 9-12 refer to the table beneath

The following table shows mortality figures for two Kebeles over a one-year period:

| Age group | Kebele A |            | Kebele B |            |
|-----------|----------|------------|----------|------------|
|           | Deaths   | Population | Deaths   | Population |
| 0-29      | 20       | 1800       | N/K      | 2200       |
| 30-59     | 80       | 2500       | N/K      | 3600       |
| 60+       | 200      | 2700       | N/K      | 1000       |
| All ages  | 300      | 7000       | 400      | 6800       |

N/K = Not Known

9. Calculate the crude overall mortality rates for Kebeles A and B: (2 marks)

Kebele A:

Kebele B:

10. What type of standardization is appropriate to compare the Kebeles, given the data above? (1 mark)

11. Use this type of standardization to produce an overall summary measure, which compares Kebele A with Kebele B and is weighted according to the age structure of each Kebele. You may wish to construct a table in the space below. (6 marks).

12. What is this summary measure called? (1 mark)

13. Which of the following is *not* found among the 'Wilson' criteria for adopting a screening program (1 mark)?

- A. The disease must be prevalent and serious
- B. The test must be simple, precise and validated
- C. The program must be acceptable to the public and health professionals
- D.** The risks of the program must outweigh the benefits
- E. Effective treatment must be available

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14. The positive predictive value of a screening test is (1 mark):
- A. The proportion of true positives correctly identified.
  - B. The proportion of true negatives correctly identified.
  - C. The proportion of true positives who are test positives.
  - D. The proportion of test positives who are true positives.
  - E. The proportion of true negatives who are test negatives.
15. In relation to time lines of infectious disease, which of the following statements is true? (1 mark)
- A. The infectious period precedes the latent period.
  - B. The symptomatic period may coincide with the incubation period.
  - C. The infectious period may coincide with the incubation period.
  - D. Recovery is always followed by immunity.
  - E. The symptomatic period precedes the incubation period.
16. In a certain population, the basic reproductive number for measles is 9, while that for rubella is 4. Which of the following statements is true? (1 mark)
- A. If one tenth of the population is immune to measles, the effective reproductive number is 10.
  - B. Assuming 100% vaccine efficacy, the fraction of the population who must be immunized to eradicate measles is 0.89 or more.
  - C. The average age of infection will be lower for rubella than for measles.
  - D. Assuming 100% vaccine efficacy, the fraction of the population who must be immunized to eradicate rubella is 0.85 or more.
  - E. If half of the population is immune to rubella, the effective reproductive number is 8.
17. The following are used to decide scientific authorship according to the 'Vancouver' criteria (1 mark):
- (i) Conception and design
  - (ii) Acquisition of data
  - (iii) Analysis and interpretation of data
  - (iv) Drafting article or revising it critically
  - (v) Giving final approval of the version submitted
- Which of the following combinations of criteria indicate eligibility for authorship (1 mark)?
- A. (i), (ii) and (iii)
  - B. (i), (iii) and (iv)
  - C. (ii), (iv) and (v)
  - D. (ii), (iii) and (v)
  - E. (i), (ii) and (iv)
18. Which of the following diseases is NOT in the Ethiopian National Priority Disease list? (1 mark)
- A. Measles
  - B. Mumps
  - C. Meningitis
  - D. Pneumonia in children less than 5 years
  - E. Malaria

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19. Which of the following is NOT a measure to manage an epidemic? (1 mark)

- A. Action to prevent transmission of disease by ingestion
- B. Active immunization
- C. Supervision
- D. Action to reduce transmission by respiratory infection
- E. Chemoprophylaxis

20. Which of the following are NOT among the objectives of Integrated Disease Surveillance and Response in Ethiopia? (1 mark)

- A. Strengthen the capacity of woredas to conduct effective surveillance activities
- B. Improve the use of information for decision making
- C. Improve laboratory capacity in identification of pathogens
- D. Emphasize hospital involvement in detection and response to public health problems
- E. Strengthen the involvement of laboratory personnel in epidemiological surveillance